

Calculating with coordinates

For ordered basis $B=(v_1, \dots, v_n)$ of V

We get isom. $\mathbb{R}^n \xrightarrow{\sim} V$

$$\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \mapsto \sum c_i v_i$$

We call α the
coordinate map
w/ basis B

$$\left\{ \begin{array}{l} [v]_B \longleftarrow v \\ \uparrow \\ \text{notation} \end{array} \right.$$

Take $\left\{ \begin{array}{l} V \text{ with basis } B=(v_1, \dots, v_n) \\ W \text{ with basis } C=(w_1, \dots, w_k) \\ \text{linear map } V \xrightarrow{f} W \end{array} \right.$

Consider the "commutative square" below of
linear maps with α, β coord maps (both isoms)

$$\begin{array}{ccc} V & \xrightarrow{f} & W \\ \alpha \downarrow & & \downarrow \beta \\ \mathbb{R}^n & \xrightarrow{\beta f \alpha^{-1}} & \mathbb{R}^k \end{array} \quad \begin{array}{ccc} v & \xrightarrow{f} & f(v) \text{ "abstract"} \\ \alpha \downarrow & & \downarrow \beta \\ [v]_B & \xrightarrow{\beta f \alpha^{-1}} & [f(v)]_C \text{ "concret"} \end{array}$$

$\downarrow \text{Coord}$

Question: Under this translation from abstract to concrete, the linear map $f: V \rightarrow W$ translates to a linear map "?" from $\mathbb{R}^n \rightarrow \mathbb{R}^k$. Now "?" must be multiplication by a unique matrix. Let's call it $M_{B,C}(f)$

We have $[f(v)]_C = M_{B,C}(f)[v]_B$, where $\begin{cases} i^{\text{th}} \text{ column of } M_{B,C}(f) = "(e_i)" \\ = [f(v)_i]_C \end{cases}$

Similarly, $[g(w)]_D = M_{C,D}(g)[w]_C$

Under the abstract $\longleftrightarrow$ concrete dictionary, we have ^{want to find this matrix}

vector $\longleftrightarrow$ column vector

linear map $\longleftrightarrow$ matrix

composition of $\longleftrightarrow$ matrix

linear maps multiplication

Let's see how: $V \xrightarrow{f} W \xrightarrow{g} U$ linear maps

Setup: $\left\{ \begin{array}{l} \text{Bases } B \quad C \quad D \end{array} \right.$

Matrices: $\left. \begin{array}{l} \text{for } f : M_{B,C}(f) \\ \text{for } g : M_{C,D}(g) \end{array} \right| \text{for } g \circ f : M_{B,D}(g \circ f)$

Note: $[g \circ f(v)]_D = M_{B,D}(g \circ f) [v]_B$

$$[g(\underline{f(v)})]_D = M_{C,D}(g) [f(v)]_C$$

$$= M_{C,D}(g) \left(M_{B,C}(f) [v]_B \right)$$

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$$\stackrel{\downarrow}{=} \left(M_{C,D}(g) M_{B,C}(f) \right) [v]_B$$

$$\therefore M_{B,D}(g \circ f) = M_{C,D}(g) M_{B,C}(f)$$

Consequences ① matrix mult is assoc b/c function composition (check details)

② Matrix inverse $\leftrightarrow$ inverse of linear map! (See exercises)

Inverse of matrix A is a matrix B st $AB=I, BA=I$

↓
• This forces A to be square

cannot be tall

cannot be wide

• For a square matrix $AB=I \Leftrightarrow BA=I$.

• A is invertible $\Leftrightarrow \text{RREF}(A)=I \Leftrightarrow A \rightarrow E_1 A \rightarrow E_2 E_1 A \rightarrow \dots \rightarrow (E_N \dots E_1) A = I$ row opⁿ.

To find A^{-1} ① row reduce A to I

② Do same seq. of row op^s to I .

To find A^{-1} just do $I \rightarrow E_1 \rightarrow E_2 E_1 \rightarrow \dots \rightarrow E_N \dots E_1 = A^{-1}$

Change of basis!

① Within a single vector space V

Two bases

$$B = (v_1, \dots, v_n)$$

$$B' = (v'_1, \dots, v'_n)$$

$$B P = B', \text{ i.e.}$$

$$B = B' P^{-1}$$

P = change of basis matrix

Express new basis B' in terms of B

$$\left. \begin{array}{l} v'_1 = P_{11} v_1 + \dots + P_{n1} v_n \\ \vdots \\ v'_n = P_{1n} v_1 + \dots + P_{nn} v_n \end{array} \right\} [v_1 \dots v_n] \overset{\text{hypervector}}{\underbrace{\begin{bmatrix} P_{11} & \dots & P_{1n} \\ \vdots & & \vdots \\ P_{n1} & \dots & P_{nn} \end{bmatrix}}} = [v'_1 \dots v'_n] \rightarrow \text{notation}$$

P = change of basis matrix

$\left\{ \begin{array}{l} i^{\text{th}} \text{ column} = \text{coord of} \\ i^{\text{th}} \text{ new basis vector} \\ \text{in terms old basis} \\ \text{vectors.} \end{array} \right.$

Question: How to get new coord in terms of old? To be done next

Claim $[v]_{B'} = P^{-1} [v]_B$

$$B[v]_B = v = B'[v]_{B'}$$

$$B'P^{-1}[v]_B = B'[v]_{B'} \Rightarrow$$

coords of same vector wrt basis B'

② Change of basis for a linear map $f: V \rightarrow W$

Similarly $[w]_{C'} = Q^{-1} [w]_C$
where Q is change of basis matrix in W from old basis C to new basis C'

$$M_{B',C'}(f) = Q^{-1} M_{B,C}(f) P$$

Try to prove this! Will see in detail in next class